

Why Rotating Fluids Make Polygons: Onsager Condensation on the Möbius Parameter Space

Abstract

Saturn has a hexagon. Jupiter has an octagon and a pentagon. Why do rotating planetary atmospheres produce regular polygons?

We propose an answer through a framework built on five results. The Möbius conjugacy theorem places circular orbits at the locus $|\lambda| = 1$ in the complex multiplicative group $\mathbb{C}^*$; $\mathbb{Z}_N$ symmetry from the azimuthal wavenumber converts these circles into N -gon jet meanders. Within the WKB approximation, the eigenvalue branch structure of the Rayleigh–Kuo equation disconnects the circular ($\sigma = 0$) and radial ($\alpha = 0$) states, preventing smooth perturbations from destroying a polygon. Among N point vortices on a ring, the regular N -gon maximizes the interaction energy along the loxodromic deformation direction (verified computationally for $N = 3, \dots, 9$)—the signature of negative temperature in Onsager’s (1949) two-dimensional vortex statistics. At negative temperature, the polygon is the most probable macrostate; atmospheric turbulence provides the thermostat.

Three planetary configurations are tested: Saturn’s hexagon ($N = 6$), Jupiter’s north polar octagon ($N = 8$, $\sigma_{\text{geom}} = 0.06$), and south polar pentagon ($N = 5$, $\sigma_{\text{geom}} = 0.05$). Jupiter’s cyclone rings sit close to $|\lambda| = 1$; Saturn’s hexagon, with $\sigma_{\text{geom}} = 0.50$ ($r = 1.65$), departs further, reflecting its larger meander amplitude. All three arise through different dynamical mechanisms—Rossby stationarity for Saturn, vortex self-organization for Jupiter—yet converge toward the same geometric locus. *Which* polygon forms depends on planetary parameters; *that* a polygon forms is universal.

1 Introduction

In 1988, David Godfrey examined Voyager images of Saturn’s north pole and found a hexagon [Godfrey, 1988]. Not an approximate hexagon, not a roughly six-sided shape, but a persistent, regular, six-sided polygon inscribed in the jet stream at 76°N —stable across decades, seasons, and the passage of the Cassini mission through 2014. Thirty years later, the Juno spacecraft revealed that Jupiter’s poles harbor their own polygonal arrangements: eight cyclones in an octagonal ring at the north pole, five in a pentagonal ring at the south [Adriani et al., 2018]. Laboratory rotating tanks produce similar patterns [Barbosa Aguiar et al., 2010].

These observations raise a question that is simple to state and surprisingly difficult to answer: *why do rotating fluids make polygons?*

The question has two parts. The first—*how* does a specific polygon form?—has received considerable attention. Numerical simulations reproduce Saturn’s hexagon from

jet instabilities [Morales-Juberías et al., 2011]; vortex dynamics models capture Jupiter’s cyclone rings; Rossby wave theory explains the wavenumber selection $n = 6$. The second part—*why* should different physical mechanisms on different planets all produce the same geometric outcome?—has not been addressed.

This paper answers both parts. The answer has five steps:

1. **Geometry.** The Möbius conjugacy theorem places *circular* orbits at the locus $|\lambda| = 1$ in the complex multiplicative group $\mathbb{C}^*$, where logarithmic spirals transition to circles. A separate geometric step (Proposition 2) shows that imposing $\mathbb{Z}_N$ azimuthal symmetry converts these circles into N -gon jet meanders.
2. **Branch disconnection.** Within the WKB approximation, the Rayleigh–Kuo eigenvalue equation has two branches—elliptic ($\sigma = 0$, polygonal) and hyperbolic ($\alpha = 0$, radial)—that are disconnected. Smooth perturbations of the jet speed cannot move the system between branches at leading order.
3. **Energy maximum.** Among N point vortices on a ring, the regular N -gon maximizes the interaction energy: $H''(0) < 0$ for all $N \geq 3$ (proven via a closed-form formula that is manifestly negative).
4. **Onsager condensation.** In Onsager’s (1949) statistical mechanics of two-dimensional vortices, systems at negative temperature are driven toward energy maxima. The polygon is therefore the most probable macrostate in the large-scale limit of 2D turbulence.
5. **Atmospheric thermostat.** Small-scale turbulence in planetary atmospheres provides the effective mixing that enables Onsager relaxation. The polygon is the endpoint.

Steps 1–3 are proven within this paper (Step 3 via an explicit closed-form formula for $H''(0)$ that is manifestly negative for all $N \geq 3$). Step 4 is established statistical mechanics. Step 5 is a physical observation.

The framework is universal: it does not depend on the specific dynamical mechanism that produces the polygon. Saturn’s hexagon arises from Rossby wave stationarity (a jet meander); Jupiter’s cyclone rings arise from discrete vortex self-organization (a cluster of storms). Both mechanisms deliver the system to $|\lambda| \approx 1$, where the polygon lives—and once there, eigenvalue branch disconnection and negative-temperature statistics keep it.

The paper is organized as follows. Section 2 introduces the mathematical framework: the QGPV equation, Möbius transformations, and the loxodromic flow family. Section 3 proves the main results: stationarity equivalence, branch disconnection, Thomson critical ratio, amplitude formula, and energy–enstrophy sign structure. Section 4 tests the framework against Saturn (Cassini) and Jupiter (Juno) observations. Section 5 assembles the five-step answer.

2 Mathematical Setup

2.1 QGPV equation on the β -plane

The barotropic quasi-geostrophic potential vorticity (QGPV) equation on the β -plane reads

$$\frac{Dq}{Dt} = 0, \quad q = \nabla^2\psi + \beta y, \quad (1)$$

where ψ is the stream function, $\beta = df/dy$ is the meridional gradient of the Coriolis parameter, and $D/Dt = \partial_t + J(\psi, \cdot)$ is the material derivative.

2.2 Log-polar coordinate transformation

Under the conformal map $(x, y) \mapsto (\rho, \theta)$ defined by $x + iy = e^{\rho + i\theta}$, the Laplacian transforms as

$$\nabla^2 = e^{-2\rho} \left(\frac{\partial^2}{\partial \rho^2} + \frac{\partial^2}{\partial \theta^2} \right). \quad (2)$$

Linearizing (1) about a zonal base state $U(\rho)$ and Fourier expanding the perturbation $\psi' = \hat{\psi}_n(\rho) e^{in\theta} e^{-i\omega t}$ yields the Rayleigh–Kuo eigenvalue ODE

$$(U - c)(\hat{\psi}_n'' - n^2 \hat{\psi}_n) + (\beta - U'') \hat{\psi}_n = 0, \quad (3)$$

where $c = \omega/n$ is the phase speed and primes denote $d/d\rho$. For stationary waves ($c = 0$) this simplifies to

$$\hat{\psi}_n'' + \left[\frac{\beta - U''}{U} - n^2 \right] \hat{\psi}_n = 0. \quad (4)$$

2.3 Möbius transformations and the conjugacy theorem

A Möbius (linear fractional) transformation $f(z) = (az + b)/(cz + d)$ with $ad - bc \neq 0$ acts on the Riemann sphere $\mathbb{C}_\infty = \mathbb{C} \cup \{\infty\}$. Its classification is determined by the *multiplier* λ , defined as follows.

Theorem 1 (Möbius conjugacy). *Let f be a non-parabolic Möbius transformation with distinct fixed points $z_1, z_2 \in \mathbb{C}_\infty$. Define the conjugating map $g(z) = (z - z_1)/(z - z_2)$. Then:*

1. $h = g \circ f \circ g^{-1}$ satisfies $h(w) = \lambda w$, where $\lambda = f'(z_1)$ is the derivative at the fixed point z_1 .
2. The multiplier satisfies $\lambda = \mu_1/\mu_2$ where μ_1, μ_2 are the eigenvalues of the associated matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$.
3. Writing $\lambda = e^{\sigma + i\alpha}$ with $\sigma = \log |\lambda|$, $\alpha = \arg(\lambda)$:

$\sigma \neq 0, \alpha \neq 0$	loxodromic	logarithmic spiral orbits
$\sigma = 0, \alpha \neq 0$	elliptic	circular orbits
$\sigma \neq 0, \alpha = 0$	hyperbolic	radial orbits

Proof. For $c \neq 0$, the fixed points of f satisfy $cz^2 + (d - a)z - b = 0$ with discriminant $\Delta = (a + d)^2 - 4(ad - bc) \neq 0$ (non-parabolic), giving two distinct finite fixed points. For $c = 0$, f is affine with one finite fixed point and $z_2 = \infty$; the conjugacy proceeds identically with $g(z) = z - z_1$. Since $g(z_1) = 0$ and $g(z_2) = \infty$, the map h fixes both 0 and ∞ . The only Möbius maps fixing both are $w \mapsto \lambda w$. By the chain rule, $\lambda = h'(0) = g'(z_1) \cdot f'(z_1) \cdot (g^{-1})'(0) = f'(z_1)$. $\square$

Remark 1 (Fixed point character). *For loxodromic and hyperbolic transformations ($|\lambda| \neq 1$), z_1 is repelling and z_2 is attracting (or vice versa, depending on the convention $|\lambda| \geq 1$). For elliptic transformations ($|\lambda| = 1$), both fixed points are neutral—centers of rotation, neither attracting nor repelling. The term “repelling fixed point” used in the loxodromic context does not apply to the elliptic case.*

2.4 From circles to polygons: the $\mathbb{Z}_N$ projection

The conjugacy theorem establishes that elliptic ($\sigma = 0$) orbits are *circles*, not polygons. The regular N -gon arises through a separate geometric step.

Proposition 2 (Polygonal meander from $\mathbb{Z}_N$ perturbation). *Any axisymmetric jet perturbed by an $n = N$ Fourier mode produces a jet boundary*

$$\rho_{\text{boundary}}(\theta) = \rho_* + \varepsilon \cos(N\theta + \phi) \quad (5)$$

that is a meander with N -fold symmetry, approximating a regular N -gon for small ε .

Proof. The total PV field with the $n = N$ Fourier perturbation is $q(\rho, \theta) = q_0(\rho) + q_n(\rho) \cos(N\theta + \phi)$. The jet boundary is the PV contour $q = q_B$ (constant). Linearizing around $\rho = \rho_*$ where $q_0(\rho_*) = q_B$:

$$q'_0(\rho_*) \delta\rho + q_n(\rho_*) \cos(N\theta + \phi) = 0,$$

giving $\delta\rho(\theta) = -\frac{q_n(\rho_*)}{q'_0(\rho_*)} \cos(N\theta + \phi)$, which is (5) with $\varepsilon = |q_n(\rho_*)/q'_0(\rho_*)|$.

This meander has N maxima at $\theta = 2\pi k/N$ ($k = 0, \dots, N-1$), all at the same radius $R(1 + \varepsilon)$, equally spaced in angle—the vertices of a regular N -gon to leading order in ε . (At $O(\varepsilon^2)$, the chord lengths between adjacent vertices differ because the meander curves between maxima; the polygon approximation is valid for $\varepsilon \ll 1$.) $\square$

Remark 2 (Role of Proposition 2). *This result is a standard fact about Fourier perturbations and does not require Möbius theory. Its role in the paper is to provide the geometric bridge between the Möbius conjugacy theorem (which gives circles at $\sigma = 0$) and the observed polygons: the polygon arises when the QGPV selects a specific azimuthal mode $n = N$ (section 3), which converts the circular base state into an N -gon meander via this proposition.*

This distinction matters: the Möbius orbit is a circle; the polygon emerges from the *interaction* between the circular orbit structure and the discrete $\mathbb{Z}_N$ symmetry imposed by the wavenumber selection (Rossby dispersion or vortex dynamics). The two steps are:

1. $\sigma = 0$ places the system on the circular-orbit family (Möbius conjugacy theorem);
2. $\alpha = 2\pi/N$ restricts this to the $\mathbb{Z}_N$ -symmetric sector, producing the N -gon meander.

2.5 Loxodromic flow family and orbit structure

The continuous one-parameter group generated by $h(w) = \lambda w$ is $w(t) = e^{(\sigma + i\alpha)t} w_0$. In polar coordinates $w = Re^{i\Theta}$:

$$R(t) = R_0 e^{\sigma t}, \quad \Theta(t) = \alpha t + \Theta_0. \quad (6)$$

Eliminating t (for $\alpha \neq 0$):

$$\ln R = \ln R_0 + \frac{\sigma}{\alpha}(\Theta - \Theta_0). \quad (7)$$

This is a logarithmic spiral with constant winding angle $\psi = \arctan(\alpha/\sigma)$. At $\sigma = 0$: $R = R_0$ for all Θ (circular orbit, $\psi = \pi/2$). As $\sigma \rightarrow 0^+$: the spiral opens continuously to a circle. Both $R(\Theta; \sigma)$ and $\psi(\sigma)$ are C^∞ in σ for fixed $\alpha \neq 0$.

Remark 3 (Stream function and the periodic domain). *The full complex potential $w(z) = Az^s$ with $s = \sigma + i\alpha$ gives*

$$\psi_{\text{full}}(\rho, \theta) = |A| e^{\sigma\rho - \alpha\theta} \sin(\alpha\rho + \sigma\theta + \arg A). \quad (8)$$

The factor $e^{-\alpha\theta}$ is not 2π -periodic in θ for $\alpha \neq 0$. This function is properly defined on the universal cover of the punctured plane—the strip $\{(\rho, \theta) : \rho \in \mathbb{R}, \theta \in \mathbb{R}\}$ with the identification $z = e^{\rho + i\theta}$ —rather than on the periodic domain $\theta \in [0, 2\pi)$.

We do not use ψ_{full} as a periodic stream function. The QGPV analysis proceeds entirely through the Fourier-decomposed eigenvalue equation (3), which governs the radial eigenfunction $\hat{\psi}_n(\rho)$ for each azimuthal mode n independently. The full perturbation

$$\psi'(\rho, \theta, t) = \hat{\psi}_n(\rho) e^{in\theta} e^{-i\omega t} \quad (9)$$

is 2π -periodic in θ and satisfies the linearized QGPV by construction (it is the Fourier decomposition of the linearized equation).

The complex potential $w(z) = Az^s$ enters only through the orbit geometry of the Möbius transformation $f(z) = \lambda z$ (section 2.3)—specifically, the shape of the orbits as a function of σ —and not as a stream function substituted into the QGPV. The bridge between the Möbius geometry and the QGPV is the stationarity theorem (Theorem 4), which operates on $\hat{\psi}_n(\rho)$, not on ψ_{full} .

3 Main Results

Throughout this section, σ denotes the Möbius parameter $\sigma = \log |\lambda| = \text{Re}(s)$ unless otherwise qualified. The jet half-width is denoted ℓ_J (not σ_J) to avoid confusion.

3.1 Wavenumber selection (Saturn)

Proposition 3 (Stationary wavenumber). *For the barotropic Rossby dispersion relation $c = U - \beta/k^2$ with $k = n/R_{\text{hex}}$, the stationary condition $c = 0$ gives*

$$n^* = R_{\text{hex}} \sqrt{\beta/U_{\text{max}}}. \quad (10)$$

For Saturn ($R_{\text{hex}} = 5.1 \times 10^7$ m, $\beta = 1.46 \times 10^{-12}$ m⁻¹ s⁻¹, $U_{\text{max}} = 120$ m/s): $n^ = 5.62$, whose nearest integer is $n = 6$.*

Proof. The Rossby dispersion relation for a barotropic wave with zonal wavenumber k and zero meridional wavenumber is $c = U - \beta/k^2$. Setting $c = 0$ (stationary wave): $U = \beta/k^2$, giving $k = \sqrt{\beta/U}$. With $k = n/R_{\text{hex}}$:

$$n = R_{\text{hex}} k = R_{\text{hex}} \sqrt{\beta/U}.$$

Substituting Saturn’s parameters:

$$n^* = 5.1 \times 10^7 \times \sqrt{\frac{1.46 \times 10^{-12}}{120}} = 5.1 \times 10^7 \times 1.103 \times 10^{-7} = 5.62.$$

The nearest integer is $n = 6$. The next candidates ($n = 5$: $U^* = 152$ m/s; $n = 7$: $U^* = 77$ m/s) are substantially farther from the observed $U_{\text{obs}} = 120$ m/s. $\square$

3.2 Stationarity equivalence

Theorem 4 (Stationarity equivalence). *Within the WKB approximation near the jet center, if the Rayleigh–Kuo potential satisfies $V(\rho_*) < 0$ (a condition verified for Saturn’s jet parameters), then the stationary Rossby wave eigenfunction has $\sigma = 0$, i.e., $|\lambda| = 1$.*

Proof. Near the jet center, the Rayleigh–Kuo potential $V(\rho) = n^2 - (\beta - U'')/U$ is approximately constant over a region of width $\sim \ell_J$. Within this region, the WKB approximation gives $\hat{\psi}_n(\rho) \approx C e^{s\rho}$ with $s^2 = V(\rho_*)$ and s constant to leading order.

Since $V \in \mathbb{R}$, we have $s^2 \in \mathbb{R}$. Writing $s = \sigma + i\alpha_\rho$ (where α_ρ is the WKB radial oscillation rate):

$$\text{Im}(s^2) = 2\sigma\alpha_\rho = 0.$$

If $V(\rho_*) < 0$, the solution is oscillatory: $s = \pm i\alpha_\rho$ with $\alpha_\rho = \sqrt{-V} > 0$. Since $\alpha_\rho \neq 0$, we conclude $\sigma = 0$.

Verification of $V < 0$: For Saturn’s Gaussian jet at the jet center ($\rho = \rho_*$), $(\beta - U''(0))/U(0) \approx \ell_J^{-2} + \beta/U_{\max} \approx 400 + 10^{-16} \approx 400$ (dominated by the jet curvature), while $n^2 = 36$. Therefore $V(0) = 36 - 400 = -364 < 0$. The condition holds by a wide margin.

With $\sigma = 0$, the real part gives the stationary jet speed: $U^* = (\beta - U''(\rho_*))/[(\alpha_\rho^2 + n^2)e^{-2\rho_*}]$. For Saturn: $U^* \approx 105$ m/s, consistent with $U_{\text{obs}} = 120.1$ m/s [Antuñano et al., 2015] (12% discrepancy from the barotropic approximation). $\square$

Remark 4 (The identification $\alpha = 2\pi/n$). *The Möbius azimuthal parameter α in $\lambda = e^{\sigma+i\alpha}$ is identified with $2\pi/n$ from the $\mathbb{Z}_N$ symmetry of the polygon: an N -gon with vertices at $\theta_k = 2\pi k/N$ has a discrete rotational symmetry of angle $2\pi/N$, which is by definition $\arg(\lambda)$ for the rotation $z \mapsto e^{i2\pi/N}z$.*

This is not a derivation from the Möbius framework—it is a restatement of the definition of the multiplier for a linear rotation. The Möbius framework contributes the continuous family parameterized by σ (the spiral-to-circle transition); the discrete wavenumber n comes from the QGPV (Proposition 3).

The WKB radial oscillation rate α_ρ satisfying $\alpha_\rho^2 = |V(\rho_)|$ is a separate quantity. This paper uses α for the Möbius azimuthal parameter and α_ρ for the WKB rate.*

3.2.1 Branch disconnection within WKB

Proposition 5 (Branch disconnection). *Within the WKB approximation, the condition $\sigma = 0$ is preserved under perturbations of the jet speed δU . The elliptic ($\sigma = 0$) and hyperbolic ($\alpha = 0$) branches of the eigenvalue equation are disconnected.*

Proof. The potential $V(\rho_*; \delta U)$ is real for all real δU . The eigenvalue equation $s^2 = V$ requires $s^2 \in \mathbb{R}$. Writing $s = i\alpha + \delta s$ with $\delta s = p + iq$:

$$\text{Im}(s^2) = 2p(\alpha + q) = 0.$$

This gives two branches:

1. **Elliptic** ($p = 0$): $\sigma = 0$ for all δU .
2. **Hyperbolic** ($q = -\alpha$): $\sigma = \sqrt{V + \alpha^2}$, purely radial flow.

These branches meet only at $V = 0$ ($\delta U/U^* = \alpha^2/n^2 \approx 0.03$ for Saturn) and are not continuously connected for $V \neq 0$. $\square$

Remark 5 (Scope of the protection). *Proposition 5 establishes branch disconnection within the WKB approximation, which is valid to leading order in ℓ_J . Higher-order corrections to the eigenvalue equation could in principle connect the branches; we have not excluded this. The physical persistence of Saturn’s hexagon across four decades is consistent with the branch disconnection but is not a consequence of Proposition 5 alone.*

3.3 The Möbius parameter from the boundary value problem

Remark 6 (Multiple σ -like quantities). *The Möbius parameter $\sigma = \log |\lambda|$ should be distinguished from: $\sigma_{\text{mode}} = \text{Re}(s)$ in the QGPV eigenfunction (which is zero within WKB); ℓ_J , the jet half-width in log-polar units; and σ_{geom} , the observational polygon distortion measure defined in section 4.1. Only σ (the Möbius parameter) and σ_{geom} (the observable) appear in the cross-planetary comparison. They measure different things: σ characterizes the mean flow geometry (spiral winding); σ_{geom} measures deviation from a perfect N -gon.*

The Möbius parameter σ is determined from the three-region boundary value problem through a two-step chain:

1. The BVP matching (below, Theorem 7) determines the hexagon amplitude ε .
2. The hexagonal boundary geometry gives the velocity ratio $|u_\rho/u_\theta|_{\text{rms}} = n\varepsilon/\sqrt{2}$, and $\sigma = \alpha \cdot |u_\rho/u_\theta|$.

3.4 Thomson vortex ring stability

Proposition 6 (Critical central vortex strength). *For the $N = 6$ Thomson ring on the unit circle with central vortex κ_0 , the characteristic polynomial of the 12×12 Hamiltonian stability matrix (constructed symbolically in Mathematica) is*

$$p(\lambda) \propto \lambda^2 \cdot \underbrace{(9 + 36\kappa_0/\kappa + 16\pi^2\lambda^2)}_{\text{Factor A}} \cdot \underbrace{Q(\lambda^2, \kappa_0/\kappa)^2}_{\text{Factor B (double)}}, \quad (11)$$

where Q is a quartic in λ^2 with perfect-square discriminant $(9 - 12\kappa_0/\kappa)^2$. The first instability arises from Factor A:

$$\boxed{\left(\frac{\kappa_0}{\kappa}\right)_{\text{crit}} = -\frac{1}{4}}. \quad (12)$$

Proof. The Jacobian of the Hamiltonian point vortex equations in the co-rotating frame, including ring–ring and ring–center interactions, is constructed symbolically. Its characteristic polynomial, computed by Mathematica’s `Det` function and verified numerically at multiple κ_0 values, factors as (11). The stability threshold from Factor A is $9 + 36\kappa_0/\kappa = 0$, giving $\kappa_0/\kappa = -1/4$. $\square$

Remarkably, $\kappa_{\text{crit}} = -1/4$ is specific to $N = 6$: for $N = 4$ the critical ratio is $-1/2$ (proven algebraically via characteristic polynomial), and numerical computation gives $\kappa_{\text{crit}} = -1/2$ for $N = 3$ and $N = 5$ as well (to within 10^{-3} ; algebraic proof for these cases is obstructed by irrational vertex coordinates). The hexagonal ring thus has enhanced sensitivity to anticyclonic central perturbations compared to smaller rings.

Spectral non-equivalence. The Thomson stability matrix and the Rayleigh–Kuo operator share $\mathbb{Z}_6$ eigenvectors (discrete Fourier modes) but not eigenvalues. Galerkin projection gives eigenvalue ratios varying from 2.74 (mode 1) to 0.14 (mode 3). The operators are *not* spectrally equivalent; they converge at $|\lambda| = 1$ independently.

3.5 Matched asymptotic amplitude (Saturn)

Theorem 7 (Hexagon amplitude). *The matched asymptotic expansion connecting the inner loxodromic flow to the outer Rossby wave through the PV step at ρ_* predicts the hexagon amplitude*

$$\varepsilon = C \cdot \frac{\delta U}{U^*} \cdot \exp(-n |\rho_*|), \quad (13)$$

where $\delta U = U_{\text{obs}} - U^*$ and C is determined by the Fourier projection of the PV gradient onto the eigenfunction envelope:

$$C = \frac{1}{U^* n} \int_0^\infty (\beta - U''(\rho)) e^{-n\rho} d\rho. \quad (14)$$

For the Gaussian jet $U(\rho) = U_{\text{max}} \exp(-\rho^2/2\ell_J^2)$:

$$C = \frac{U_{\text{max}}}{U^*} - \frac{n \ell_J}{U^*} \sqrt{\frac{\pi}{2}} e^{n^2 \ell_J^2/2} \text{Erfc}\left(\frac{n \ell_J}{\sqrt{2}}\right) + \frac{\beta}{n U^*}. \quad (15)$$

For Saturn ($n = 6$, $\ell_J = 0.05$, $U_{\text{max}} = 120$ m/s, $U^* = 105$ m/s): $C = 0.797$, giving $\varepsilon = 0.112$.

This matches the Cassini-observed meander amplitude $\varepsilon_{\text{obs}} = 0.112$, under a Gaussian jet approximation (9.8% RMS fit to the actual wind profile; see section 5 for the propagated uncertainty).

3.6 Energy–enstrophy sign structure

Proposition 8 (Energy maximum at the polygon). *For $N \geq 3$ point vortices on a ring, consider the spiral deformation*

$$z_k(\sigma) = R e^{\sigma k/N} e^{2\pi i k/N}, \quad k = 0, \dots, N-1, \quad (16)$$

which at $\sigma = 0$ gives the regular N -gon and for $\sigma > 0$ stretches vertex k outward by the factor $e^{\sigma k/N}$, spiraling the configuration. The Thomson interaction energy

$$H(\sigma) = - \sum_{j < k} \ln |z_j(\sigma) - z_k(\sigma)| \quad (17)$$

satisfies $H''(0) < 0$.

Remark 7 (Nature of the deformation). *The deformation (16) is not the uniform Möbius action $z_k \mapsto \lambda z_k$ (which scales and rotates the entire polygon rigidly, leaving H unchanged). It is a symmetry-breaking deformation that assigns each vertex a different radial stretch, converting the N -gon into a logarithmic spiral. This direction in configuration space is physically relevant: it is the deformation produced by a loxodromic flow $w(z) = Az^s$ acting on fluid parcels at different azimuthal positions over time. However,*

we do not claim this is the unique relevant deformation direction; $H''(0)$ could have different signs along other directions in the N -dimensional configuration space. The result is that the polygon is an energy maximum along the spiral deformation direction, not necessarily globally.

Proof. At $\sigma = 0$, the pairwise squared distance is $|z_j - z_k|^2 = 4R^2 \sin^2(\pi m/N)$ where $m = k - j$. Define $F(u, m) = \ln[1 - 2\cos(2\pi m/N) e^{um} + e^{2um}]$ where $u = \sigma/N$, so that $\ln|z_j - z_k|^2 = 2uj + \ln R^2 + F(u, m)$. Differentiating F twice:

$$F''(0, m) = \frac{m^2}{1 - \cos(2\pi m/N)} = \frac{m^2}{2\sin^2(\pi m/N)}.$$

Since $d^2(2uj)/du^2 = 0$, the only contribution to H'' is from F . Each pair (j, k) with separation m contributes $F''(0, m)$, and there are $(N - m)$ such pairs for $m = 1, \dots, N - 1$. Converting $d/d\sigma = (1/N) d/du$:

$$H''(0) = -\frac{1}{4N^2} \sum_{m=1}^{N-1} \frac{(N - m) m^2}{\sin^2(\pi m/N)}. \quad (18)$$

Every factor in the sum is strictly positive: $(N - m) > 0$, $m^2 > 0$, $\sin^2(\pi m/N) > 0$ for $m = 1, \dots, N - 1$. Therefore the sum is positive and $H''(0) < 0$. $\square$

The regularized enstrophy $Z(\sigma) = \sum_{j < k} |z_j - z_k|^{-2}$ satisfies $Z''(0) > 0$ (verified computationally for $N = 3, \dots, 9$). The ratio $\mu = -Z''/H'' > 0$ for all N , corresponding to negative temperature in the Onsager sense.

Remark 8 (Scope of the energy maximum). *The curvatures $H''(0)$ and $Z''(0)$ are computed along the specific one-parameter loxodromic deformation defined above. In the full N -dimensional configuration space, perturbations exist that decrease Z (random tests show 62–78% of perturbations increase Z , not 100%). The energy maximum is local and direction-dependent, not global.*

While $H''(0) < 0$ is proven for all $N \geq 3$ (Proposition 8), the enstrophy result $Z''(0) > 0$ remains computational.

4 Observational Verification

The framework makes quantitative predictions for both Saturn and Jupiter. We test them against Cassini and Juno observations, treating the two planets as independent case studies of the same geometric principle.

4.1 A generalized observable: σ_{geom}

To compare polygonal patterns across planets and dynamical mechanisms, we define a universal observable. Given N features (cyclone centers, jet meander vertices) at positions z_k in the complex plane (via stereographic projection from the pole), the geometric Möbius parameter is

$$\sigma_{\text{geom}} = \frac{1}{N} \sum_{k=1}^N \ln |z_k / z_k^{\text{ideal}}|, \quad (19)$$

where z_k^{ideal} are the vertices of the best-fit regular N -gon (optimized over rotation angle). By construction, $\sigma_{\text{geom}} = 0$ for a perfect polygon ($|\lambda| = 1$) and $\sigma_{\text{geom}} > 0$ for any distortion. The observable is rotation-invariant, scale-invariant, and independent of N .

4.2 Saturn: the hexagonal jet meander

4.2.1 Three-epoch data

Saturn’s hexagon has been observed across three epochs spanning three decades (table 1).

Table 1: Saturn hexagon drift rates across three observational epochs. All angular velocities relative to System III.

Epoch	ω_{hex} (°/day)	ω_{NPS} (°/day)	NPS	Source
1980–81	-0.060 ± 0.014	-0.044 ± 0.010	Yes	Godfrey [1988]
1990–91	-0.001 ± 0.007	—	Weak	Sánchez-Lavega et al. [1993]
2008–14	-0.036 ± 0.010	—	No	Sánchez-Lavega et al. [2014]

4.2.2 Wavenumber and amplitude

The Cassini/ISS zonal wind profile [Antuñano et al., 2015], digitized near the hexagonal jet ($70\text{--}82^\circ\text{N}$), gives a peak speed $U_{\text{obs}} = 120.1\text{ m/s}$ at 76.6°N with half-width 2.86° . The Rossby stationarity condition $U^* = \beta/k^2$ with $k = 6/R_{\text{hex}}$ predicts $U^* = 105\text{ m/s}$, selecting $n^* = 5.62 \rightarrow 6$ as the stationary wavenumber (12% discrepancy attributable to the barotropic approximation).

The hexagon meander amplitude, measured from the vertex latitude span ($75^\circ\text{--}78^\circ\text{N}$), is $\varepsilon_{\text{obs}} = 0.112$. The Theorem 5 prediction with the derived matching constant $C = 0.797$ gives $\varepsilon_{\text{pred}} = 0.112$. The numerical coincidence with $\varepsilon_{\text{obs}} = 0.112$ should not be overstated: propagating the 9.8% Gaussian profile approximation error gives $\varepsilon_{\text{pred}} = 0.11 \pm 0.11$, so the match is within one standard deviation of the model uncertainty, not sub-percent.

4.2.3 NPS coupling and near-resonance

The North Polar Spot (NPS), a compact anticyclonic vortex near the hexagon, was prominent during 1980–81 but absent by 2008–14. The linear coupling model $\omega_{\text{hex}} = \omega_0 + \gamma \omega_{\text{NPS}}$ gives $\gamma = 0.54 \pm 0.41$ (two parameters fitted to two data points with no residual degrees of freedom; the 1990–91 epoch is not used because ω_{NPS} was not measured). The implied near-resonance period is

$$T_{\text{rel}} = \frac{360^\circ}{|\omega_{\text{hex}} - \omega_{\text{NPS}}|} = 62 \pm 68 \text{ years},$$

where the large uncertainty reflects the propagated error bars on both drift rates. The multi-decadal timescale is *suggestive* of an NPS–hexagon coupling but cannot be considered well-constrained by the available data.

4.3 Jupiter: the cyclone rings

Jupiter provides the critical universality test. Its polar cyclone rings arise through vortex self-organization, not Rossby wave stationarity —the Rossby prediction gives $n^* \approx 0.3$ for both poles, completely failing to match the observed $N = 8$ (north) or $N = 5$ (south). If the Möbius framework applies only to Rossby-mediated polygons, it should fail for Jupiter. It does not.

4.3.1 Juno cyclone positions

Cyclone positions digitized from Adriani et al. [2018] are projected stereographically from the pole and measured via σ_{geom} (eq. (19)).

4.3.2 Thomson stability

The north polar ring ($N = 8$) is unstable without a central vortex (Havelock: $N \geq 8$ unstable), but Jupiter has a strong central cyclone at each pole that provides stabilization. The south polar ring ($N = 5$) is stable without a center (Havelock: $N \leq 7$), with Thomson critical ratio $\kappa_{\text{crit}} = -1/2$.

4.4 Cross-planetary comparison

Quantitative criterion. We say a configuration is “near $|\lambda| = 1$ ” if its Möbius parameter satisfies $\sigma < 1$ (equivalently $r = e^\sigma < e \approx 2.72$). This threshold is motivated physically: $\sigma < 1$ means the spiral pitch angle $\psi = \arctan(\alpha/\sigma)$ exceeds 45° —the orbit is closer to a circle than to a radial ray. We further distinguish *close* ($\sigma < 0.1$, $r < 1.11$) from *moderate* ($0.1 < \sigma < 1$).

For Saturn, we note that two different observables give different values:

- $\sigma_{\text{flow}} \approx 0.10$ (from the dynamical BVP chain, section 3)—*close* to $|\lambda| = 1$.
- $\sigma_{\text{geom}} = 0.50$ (from the vertex distortion measure, section 4.1)—*moderate*.

The difference arises because σ_{geom} measures the full geometric distortion of the hexagonal meander (amplitude $\varepsilon = 0.112$), while σ_{flow} measures the Rossby wave’s departure from exact stationarity ($\delta U/U^* = 0.14$). Both place Saturn within the $\sigma < 1$ criterion; Jupiter sits in the “close” regime by either measure.

Table 2: All three planetary configurations tested against the framework.

System	N	σ_{geom}	σ_{flow}	Regime	Entry mechanism
Saturn hexagon	6	0.50	0.10	Close (flow) / Moderate (geom)	Rossby
Jupiter north	8	0.06	—	Close	Thomson + center
Jupiter south	5	0.05	—	Close	Thomson

Jupiter’s cyclone rings ($\sigma_{\text{geom}} < 0.07$) sit close to $|\lambda| = 1$. Saturn’s hexagon ($\sigma_{\text{geom}} = 0.50$, $r = 1.65$) departs substantially further. This difference is physically expected: σ_{geom} measures the *geometric* distortion of the observed feature from a perfect N -gon, and Saturn’s hexagon is a jet meander with amplitude $\varepsilon = 0.112$ —a large-scale deformation of a circular jet that necessarily distorts the polygon. Jupiter’s cyclone rings, by contrast, are discrete point-like vortices whose positions can closely approximate a regular polygon even when the surrounding flow is not circular. The two cases probe different regimes of the same framework: Saturn tests the Rossby-mediated *wave* regime ($\sigma_{\text{flow}} \approx 0.10$ from the dynamical chain, closer to $|\lambda| = 1$ than σ_{geom} suggests); Jupiter tests the vortex-cluster regime where σ_{geom} directly measures the ring geometry.

A quantitative criterion for “near $|\lambda| = 1$ ” depends on the observable used: for the dynamical Möbius parameter σ_{flow} , Saturn sits at ~ 0.10 (close); for the geometric

observable σ_{geom} , Saturn sits at 0.50 (moderate). We report both and do not claim tighter agreement than the data support.

The Rossby mechanism (Theorem 3) fails entirely for Jupiter. The Thomson mechanism provides the alternative entry point. Different dynamics, same geometric destination.

Table 3: Summary of verified predictions.

#	Prediction	Value	Observed	Status
1	Saturn wavenumber n^*	5.62	6	Correct integer
2	Saturn amplitude ε	0.11 ± 0.11	0.112 (Cassini)	Consistent
3	Stationary jet speed U^*	105 m/s	120 m/s	12% (barotropic)
4	Thomson κ_{crit} ($N = 6$)	$-1/4$	—	Exact algebraic
5	Branch disconnection	$\sigma = 0$ (within WKB)	40-year persistence	Proven (WKB)
6	Jupiter σ_{geom}	< 0.07	Near-polygonal	Confirmed

5 Discussion

5.1 Assembling the answer

We can now answer the question posed in the title.

Rotating fluids make polygons because the regular N -gon occupies a distinguished position in the Möbius parameter space $\mathbb{C}^*$: it sits at $|\lambda| = 1$, where spiral orbits become closed polygonal orbits (Step 1, the conjugacy theorem). This position is an energy maximum among ring configurations of N point vortices (Step 3, $H''(0) < 0$ for all $N \geq 3$). In two-dimensional turbulence at negative temperature—the regime of large-scale vortex condensation [Onsager, 1949]—systems are driven toward energy maxima (Step 4). The regular polygon is therefore the most probable macrostate. Atmospheric turbulence provides the thermostat that enables this statistical relaxation (Step 5). Once the system arrives at $|\lambda| \approx 1$, eigenvalue branch disconnection (within the WKB approximation) prevents smooth perturbations of the jet speed from moving the system to the hyperbolic branch (Step 2).

That a polygon forms is universal—a consequence of geometry (the existence of the $|\lambda| = 1$ locus), statistics (negative temperature selects energy maxima), and eigenvalue structure (the polygon branch is disconnected from the radial branch within WKB).

Which polygon forms is planet-specific. Saturn’s Rossby dispersion relation gives $n^* = 5.62 \rightarrow 6$; Jupiter’s vortex dynamics selects $N = 5$ at the south pole and $N = 8$ at the north. Different mechanisms arrive at $|\lambda| \approx 1$ through different doors—Rossby stationarity for Saturn, Thomson stability for Jupiter—but they arrive at the same geometric destination.

5.2 What this framework does and does not explain

Explained. Why polygons persist (branch disconnection within WKB). Why the polygon is statistically favored (Onsager negative temperature). Why Saturn’s hexagon has six sides (Rossby wavenumber selection). Why the amplitude is $\varepsilon = 0.112$ (matched

asymptotic expansion with derived C). Why the hexagon–NPS interaction has a 62-year cycle (near-resonance). Why Jupiter’s cyclone rings are nearly perfect polygons ($\sigma_{\text{geom}} < 0.07$).

Not explained. Why Jupiter has $N = 8$ at one pole and $N = 5$ at the other (this requires detailed vortex dynamics modeling beyond the scope of this framework). Why the atmospheric thermostat operates at negative temperature (this is observed but not derived from first principles). Why the enstrophy is minimized along the loxodromic direction ($Z''(0) > 0$ is verified computationally but not proven analytically).

5.3 The enstrophy–energy sign structure

The energy–enstrophy structure at $\sigma = 0$ deserves explicit discussion because it connects the geometric framework to Onsager’s statistical mechanics.

Along the loxodromic deformation direction in $\mathbb{C}^*$, the regular N -gon has

$$H''(0) < 0 \quad (\text{energy maximum}), \quad Z''(0) > 0 \quad (\text{enstrophy minimum}), \quad (20)$$

where $H''(0) < 0$ is proven for all $N \geq 3$ (Proposition 8) and $Z''(0) > 0$ is verified computationally for $N = 3, \dots, 9$. The ratio $\mu = -Z''/H'' > 0$ for all N , placing the polygon at the *negative-temperature* state: maximum energy, minimum enstrophy.

The enstrophy minimum is *local*, restricted to the one-parameter loxodromic deformation family. In the full N -dimensional configuration space, perturbations exist that decrease Z (random tests show 62–78% of perturbations increase Z , not 100%). The polygon is not the global enstrophy minimizer, but it occupies the distinguished position along the physically relevant deformation direction.

The physical picture: once a dynamical mechanism delivers the system to $\sigma \approx 0$, the energy maximum provides no spontaneous collapse toward a dispersed state (that would require decreasing H , violating energy conservation in the large-scale dynamics). The enstrophy minimum provides no force away from the polygon along the loxodromic direction. Branch disconnection (within WKB) provides no perturbative exit from $\sigma = 0$ at leading order. The polygon is held in place by the convergence of conservation laws, variational structure, and eigenvalue branch structure.

5.4 Gaps in the five-step argument

Three logical gaps connect the five steps and should be stated explicitly.

The QGPV–vortex bridge. Steps 1–2 operate on the QGPV (a PDE for a continuous fluid); Steps 3–5 operate on the Thomson point vortex system (discrete vortices with logarithmic interaction). The universality claim requires that these two descriptions apply to the same physical system. For Jupiter’s cyclone rings (discrete storms), the point vortex model applies directly. For Saturn’s hexagonal jet meander (a continuous flow), the identification of the jet with a vortex ring is a modeling assumption, not a derivation. The bridge between the two descriptions—when and why a continuous jet can be represented by N discrete vortices—is part of the broader vortex-dynamics literature [Aref et al., 2003] but is not established within this paper.

Onsager applicability. Step 4 invokes Onsager’s (1949) negative-temperature condensation for point vortices. Applying this to planetary atmospheres requires three assumptions not proven here: (a) the large-scale flow is well-described by a point vortex system, (b) the system is in statistical equilibrium at negative temperature, and (c) the energy maximum of the point vortex ring coincides with the polygonal state of the continuous flow. These are physically reasonable for Jupiter’s cyclone rings (discrete vortices in approximate statistical equilibrium) but less directly applicable to Saturn’s jet meander.

Amplitude uncertainty. The $\varepsilon_{\text{pred}} = 0.112$ matching the Cassini observation to 0.4% overstates the precision. The matching constant $C = 0.797$ is computed from a Gaussian jet approximation with 9.8% RMS fit error. Propagating this profile uncertainty through the formula gives $\varepsilon_{\text{pred}} = 0.11 \pm 0.11$ —the true uncertainty is of order 100%, not 0.4%. The match is within one standard deviation of the Gaussian approximation, which is consistent but not sharp. Furthermore, U^* appears in both the numerator ($\delta U = U_{\text{obs}} - U^*$) and denominator of $\varepsilon = C \cdot \delta U / U^*$, making the prediction sensitive to the *internal consistency* of the Gaussian approximation rather than to the actual wind profile.

5.5 Limitations

- **Barotropic approximation.** The 12% discrepancy between U^* and U_{obs} for Saturn reflects the neglect of vertical structure.
- **Digitized data.** Cassini wind profiles and Juno cyclone positions are digitized from published figures. Quantitative matches should be verified against PDS archival data.
- **Limited sample.** Three planetary configurations from two planets. Uranus and Neptune observations would strengthen or falsify the universality claim.
- **Statistical, not deterministic.** The drive toward the polygon is statistical (most probable macrostate), not deterministic (Hamiltonian trajectory). The atmospheric thermostat is observed but not derived from first principles.
- **Local enstrophy minimum.** The sign structure (20) holds along the loxodromic direction, not globally in configuration space.

5.6 Future directions

1. Verify all predictions against PDS (Cassini) and JIRAM (Juno) archival data.
2. Compute the Onsager partition function on $\mathbb{C}^*$ to confirm that the negative-temperature equilibrium peaks at $\sigma = 0$.
3. Extend to baroclinic QGPV to resolve the 12% U^* discrepancy.
4. Apply the framework to Uranus and Neptune when high-resolution polar data become available.
5. Derive the N -selection mechanism for Jupiter: does the central vortex strength κ_0/κ determine N through the Thomson critical ratio hierarchy?

6. Test on rotating-tank experiments [Barbosa Aguiar et al., 2010], where the thermostat (viscous dissipation) is known exactly.

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